ROUTEIQ

Smart Bus Management System for Safer and Sustainable College Transport

Theme: Sustainable Cities

1. Title of the Project / Product

ROUTEIQ: A Smart Bus Management System for College Transportation

2. Problem Statement

In many colleges, bus transport is still managed using manual registers, phone calls, and separate Excel files. Students usually wait at their stops without knowing where the bus is or whether it is delayed. Sometimes they reach the stop too early and wait for a long time, while at other times they may miss the bus because they did not receive proper information.

The transport department also faces problems in managing buses. It is not easy to know the live location of every bus, how many students are travelling in it, or how many students get on and off at each stop. Fee details are also maintained separately, so it takes time to check whether a student using the bus has paid the transport fee.

Because of these issues, buses may become overcrowded on some routes and underused on others. This can lead to poor route planning, unnecessary fuel use, and more work for the transport staff.

3. Solution Proposed

Our team is proposing **ROUTEIQ**, a smart bus management system that brings bus tracking, passenger details, route information, and transport-fee status into one web application.

A small device will be placed in the bus. It will use an ESP32 controller, GPS module, GSM/4G module, and IR/ToF sensors. The GPS module will collect the live bus location. The sensors will help count students entering and leaving the bus. This data will be sent to the server and shown on the web dashboard.

Students can check their bus location and approximate arrival time. The driver can update trip information if required. The transport administrator can monitor all buses, routes, passenger counts, and fee status from one place.

Main Features

- Live location tracking of buses.
- Estimated arrival time at bus stops.
- Passenger count at every stop.
- Bus occupancy information.
- Student, route, bus, and stop details.
- Transport-fee paid and pending status.

- Route-wise and campus-wise reports.
- Alerts for delays, breakdowns, and overcrowding.
- Admin dashboard for transport management.

4. Implementation Plan

Our team will first understand how the existing college transport system works. We will collect basic details such as the number of buses, routes, stops, and students using college transport.

After that, we will build a basic hardware prototype using ESP32, GPS, GSM/4G, and passenger sensors. The GPS module will send bus location data, while the sensors will record the number of people entering and leaving the bus.

For the software part, we will create the web pages using HTML, CSS, JavaScript, and Bootstrap for the frontend. Python FastAPI will be used for the backend. PostgreSQL will be used to store student details, bus details, routes, stops, fee information, and passenger records.

We will initially test the project on one or two buses. After checking the GPS accuracy, sensor readings, and dashboard output, we will make improvements. Later, the same system can be used for more buses and routes.

5. Safety and Efficiency Analysis

ROUTEIQ can improve safety because students and transport staff can see the bus location in real time. If a bus is delayed or travels outside its planned route, the administrator can be informed. Occupancy information can also help the college identify buses that are too crowded.

The system can reduce manual work for the transport department. Instead of checking different registers or contacting drivers repeatedly, staff can see important details on one dashboard. They can know which buses are active, which routes are busy, and how many students use each stop.

The system can also help improve efficiency. If the college identifies routes with fewer passengers, it can change the bus schedule or use a smaller bus. This can reduce unnecessary trips, save fuel, and support sustainable transportation.

6. Scalability and Future Development

For the first stage, our team plans to use ROUTEIQ for around **10 buses and 10 routes**. The system can later be expanded based on the college's requirements.

- QR code or RFID-based student bus passes.
- Online transport-fee payment.
- Android mobile application.
- AI camera-based passenger counting.
- More accurate bus arrival-time prediction.
- Automatic route suggestions based on passenger data.
- Overspeed and driver-behaviour alerts.

- College ERP integration.
- Support for electric college buses.

7. Innovation Advantage

Most existing bus-tracking systems mainly show the location of a bus. ROUTEIQ is different because it combines several useful features in one platform. Along with live tracking, it also provides passenger counting, stop-wise data, fee-status monitoring, route reports, and campus-wise information.

The project is planned as a low-cost solution. Instead of using expensive commercial systems, we are using affordable components such as ESP32, GPS modules, GSM/4G modules, and simple sensors. We are also using web technologies that are easier for students to develop and maintain.

ROUTEIQ helps the college answer more than just “Where is the bus?” It can also show how many students are in the bus, which stops are busy, which routes need improvement, and how many students have paid their transport fee.

8. Visual Impact and User Journey

Bus starts trip

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GPS tracks bus location

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Students checks bus current location

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Sensor record passenger data

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Passenger count at each stop & campus wise student count

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Data is sent to the ROUTEIQ server

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Data is processed and stored

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Check bus for breakdown or fault

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Transport staff use reports to improve bus routes

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Centralized Admin Dashboard

Before ROUTEIQ

A student waits at the bus stop without knowing the bus location. The transport staff may need to call the driver to check the bus status. Passenger and fee information may be written in separate registers. This takes time and can lead to mistakes.

After ROUTEIQ

The student opens the ROUTEIQ web application and checks the live bus location. The student can see the approximate arrival time. The bus device sends GPS and passenger data to the server. The transport administrator checks the dashboard to view bus location, passenger count, route details, and fee information.

ROUTEIQ can make college transport easier for students and simpler for transport staff. It can help reduce waiting time, improve bus usage, reduce manual work, and support better route planning.